

Portfolio #1. Concept: Newton's Second Law of Motion Calculator. This is a calculator in determining the mass, acceleration, or force of a given matter.

PROBLEM 1: Calculating Mass (m) Scenario: You are helping move furniture, and you push a heavy crate with a force of 150 N. The crate accelerates across the floor at a rate of 3 m/s². What is the mass of the crate?

PROBLEM #2: Calculating Acceleration (a) Scenario: A professional bowler throws a 7 kg bowling ball down the lane. They exert a forward force of 35 N on the ball at the moment of release. How fast does the ball accelerate?

PROBLEM #3: Calculating Force (F) Scenario: An engineer is testing a new motorized cart that has a total mass of 200kg. The design requirements state that the cart must be able to accelerate from a standstill at a rate of 1.5 m/s². How much force do it need to produce?

```
In [13]: # Computing for Mass
# This sets up the variable Force and its given value for the value.
Force = float(input("Force (N):"))

# This sets up the variable Acceleration and its given value for the value.
Acceleration = float(input("Acceleration (m/s^2):"))

# This section complete the instructions given by the code and executes it.
Mass=float(Force/Acceleration)
# This displays the result to the user.
print("The mass of the crate is", Mass, "kg.")
```

The mass of the crate is 50.0 kg.

```
In [14]: # Computing for Acceleration
# This sets up the variable Force and its given value for the value.
Force = float(input("Force (N):"))

# This sets up the variable Mass and its given value for the value.
Mass = float(input("Mass (kg):"))

# This section complete the instructions given by the code and executes it.
Acceleration=float(Force/Mass)
# This displays the result to the user.
print("The ball accelerates at a rate of", Acceleration, "m/s^2.")
```

The ball accelerates at a rate of 5.0 m/s².

```
In [16]: # Computing for Force
# This sets up the variable Acceleration and its given value for the value.
Acceleration = float(input("Acceleration (m/s^2):"))

# This sets up the variable Mass and its given value for the value.
Mass = float(input("Mass (kg):"))

# This section complete the instructions given by the code and executes it.
Acceleration=float(Mass*Acceleration)
```

```
# This displays the result to the user.  
print("The amount of force that the motor needs to produce is", Force, "N.")
```

The amount of force that the motor needs to produce is 35.0 N.